National University of Computer and Emerging Sciences
Lahore Campus

Linear Algebra (MT1004)

Sessional-II Exam

Date: April 5th 2024
Course Instructor(s)
Dr. Hira Iqbal

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 2

Roll No ______ Section ______ Student Signature ______

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Attempt all the questions.

CLO # 2: Recognize vector spaces and/or subspaces and develop their bases.

Q1: [10 marks]

a) Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $u = (u_1, u_2)$ and $v = (v_1, v_2)$:
$$u + v = (u_1 + v_1 + 2, \quad u_2 + v_2 + 2), \quad ku = (ku_1, ku_2)$$
Identify the zero vector of the set.

b) Consider the set $\{1, x + x^2, x^3\}$. Determine whether the set spans P_3. Does the set form a linearly independent set in P_3. (Solve by inspection. No calculation is required.)

CLO 2#: Recognize vector spaces and/or subspaces and develop their bases.

Q2: [20 marks]

a) Let S be the set of all vectors of the form $(2t, 0, -t)$. Determine whether the set forms a subspace in R^3.

b) Express $(-9, -7, -15)$ as a linear combination of $u = (2, 1, 4), v = (1, -1, 3)$, and $w = (3, 2, 5)$.

c) The vectors $v_1 = (1, -2, 3)$ and $v_2 = (0, 5, -3)$ are linearly independent. Enlarge $\{v_1, v_2\}$ to a basis for R^3.

d) Obtain a basis for the solution space of the homogeneous linear system, and compute the dimension of the space.

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 0 \\ 2x_1 - 6x_2 + 2x_3 &= 0 \\ 3x_1 - 9x_2 + 3x_3 &= 0 \end{aligned}$$